



ME 300:
Engineering Thermodynamics

Ideal Gas Equation
of State: Calorific

Energy, enthalpy, specific heat

$$U \sim [J], \quad u \sim [J/kg], \quad \bar{u} \sim [J/mol]$$

$$H = U + PV \sim [J]$$

$$h = u + Pv \sim [J/kg]$$

$$\bar{h} = \bar{u} + P\bar{v} \sim [J/mol]$$

$$C_v = \left(\frac{\partial u}{\partial T} \right)_v \sim [J/kg \cdot K]$$

$$C_p = \left(\frac{\partial h}{\partial T} \right)_p$$

Calorific equations of state

$$\text{S.C.S.} \rightarrow u = u(T, v) \\ u = u(T)$$

$$\text{If } \underline{\text{ideal gas}} \rightarrow u = u(T) \\ h = h(T)$$

$$\Rightarrow c_v = \left(\frac{\partial u}{\partial T} \right)_v = \frac{du}{dT} \rightarrow du = c_v dT \\ du = c_v(T) dT \quad \nearrow \text{"calorically perfect gas"} \\ \Delta u = \int_{T_1}^{T_2} c_v(T) dT \xrightarrow{c_v = \text{const.}} \Delta u = \underline{\underline{c_v \Delta T}}$$

$$\Rightarrow c_p = \left(\frac{\partial h}{\partial T} \right)_p = \frac{dh}{dT} \rightarrow dh = c_p dT \\ dh = c_p(T) dT \\ \Delta h = \int_{T_1}^{T_2} c_p(T) dT \xrightarrow{c_p = \text{const.}} \Delta h = \underline{\underline{c_p \Delta T}}$$